

Lecture 3

Memory Organization

❖ 8051 Microcontroller Memory

- ❑ 8051 can have four different memories:

- Internal ROM {4KB}
- Internal RAM {128 byte}
- External ROM {Max 64KB}
- External RAM {Max 64KB}

❖ Architecture of 8051 Microcontroller

- ❑ 8051 follows Harvard Architecture.

- ❑ So, with 8051 Program memory is ROM and Data memory is RAM.

Memory Organization

❖ Applications of 8051 Microcontroller

- ☐ Microcontroller is used in many embedded system applications. {Remote control, Microwave Oven, Washing Machine etc.}
- ☐ Once controller is embedded in application, program of controller should be fixed and it should be stored in ROM.
- ☐ Data may change in application like time, temperature and it will be stored in RAM.
- ☐ Fixed data will be stored in ROM as look up table. {ASCII, SSD etc.}

ROM Organization

Only Internal ROM

$$\overline{EA} = 1$$



Internal & External ROM

$$\overline{EA} = 1$$



Only External ROM

$$\overline{EA} = 0$$



Internal RAM Structure

❖ 8051 Microcontroller Register Bank in RAM

- ❑ 8051 has four register banks. Each register bank has Eight registers R0-R7.
- ❑ Selection of register bank can be done by two bits of PSW register, by PSW.3 and PSW.4 we can select any register bank.

CLR PSW.4

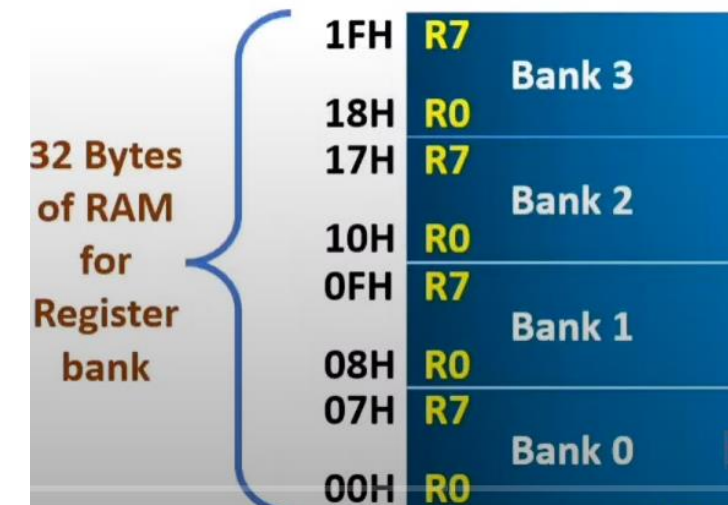
SETB PSW.3 ;Here RS = 01 means bank 1 is selected

- ❑ RAM address 00H to 1FH holds four register banks.

MOV A,R0 ;Copy R0 into A

MOV A,08H ;Copy R0 of register bank 1 into A

RS1	RS0	Register Bank
0	0	0
0	1	1
1	0	2
1	1	3

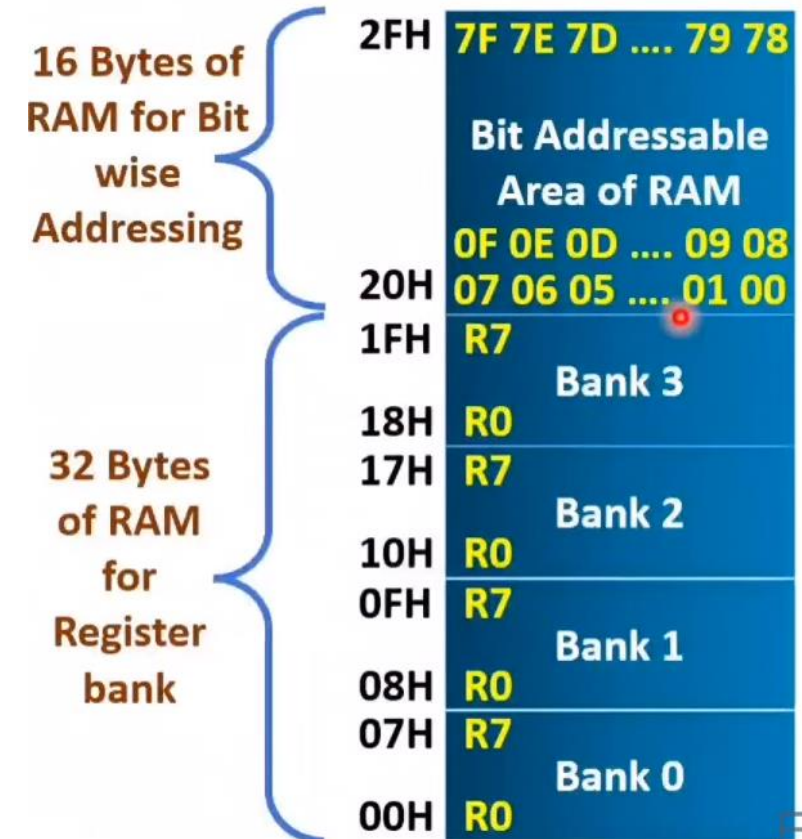


Internal RAM Structure

❖ 8051 Microcontroller Bit Addressable Area in RAM

- ❑ 16 Byte of RAM from 20H to 2FH holds bit addressable area in internal RAM of 8051 microcontroller.
- ❑ $16 \times 8 = 128$ Addressable bits in these area.
- ❑ These bits are addressed as per 00H to 7FH, in total 128 bits.
- ❑ These locations can have bit as well byte wise operations.

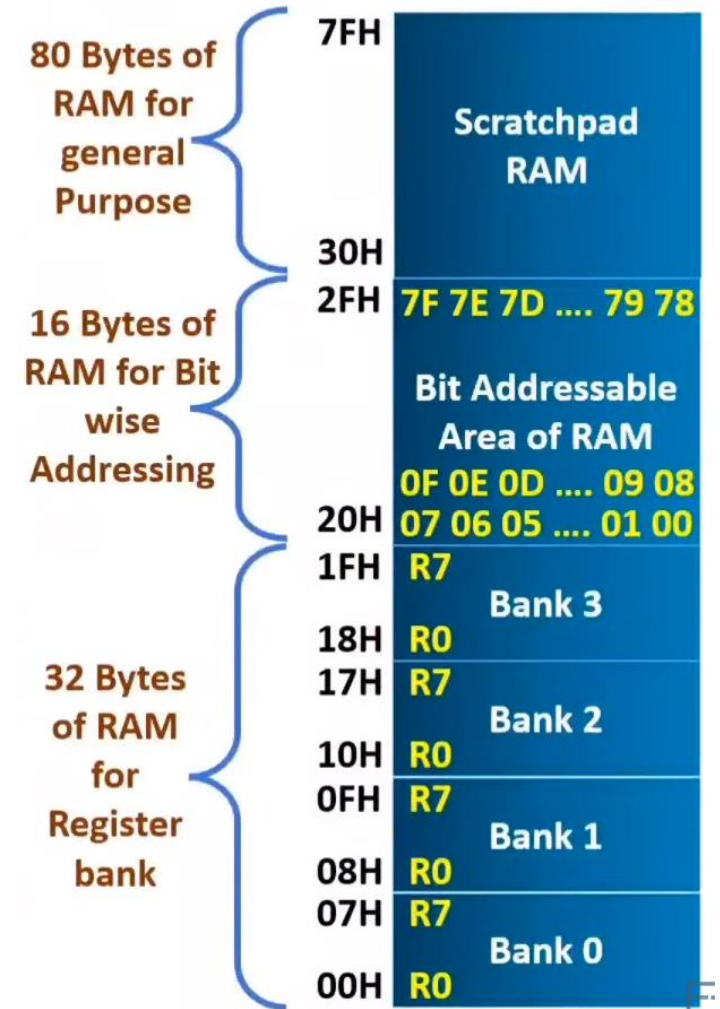
SETB 7FH ;Set MSB of 2FH RAM location
CLR 08H ;Clear LSB of 21H RAM location
MOV 20H, #FFH ;Set all bits 20H RAM locations



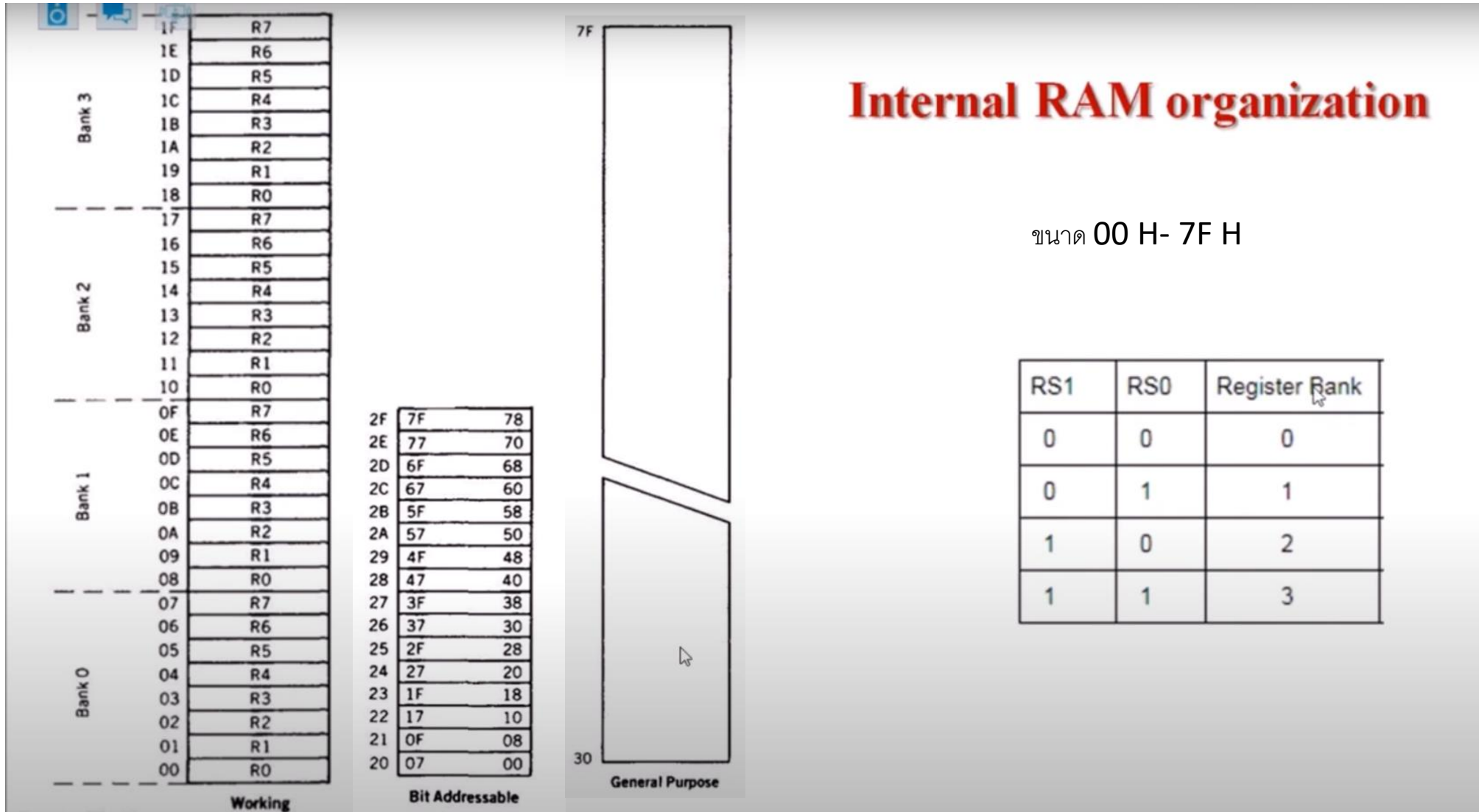
Internal RAM Structure

❖ 8051 Microcontroller General Purpose Area in RAM {Scratchpad RAM}

- ❑ 80 Byte of RAM from 30H to 7FH holds general purpose area in internal RAM of 8051 microcontroller.
- ❑ It can be used for general purpose operations.



Internal Memory Organization



Stack of 8051

- Stack of 8051 operating with respect to only memory of **Internal RAM** of 8051 microcontroller.
- Stack memory works as per LIFO {Last In First Out}.
- Stack is used to store return address during ISRs and Subroutines.
- Stack is also used by programmer using PUSH and POP instructions.
- Top of stack is pointed by SP register.
- SP is 8 bits register.
- On RESET of 8051, SP holds 07H address.
- Programmer can change the SP address as per their requirement. Range is available from 00H to 7FH as internal RAM size is of 128 bytes.

MOV SP,#20H ;SP holds 20H address of RAM

- Program to access stack memory using PUSH and POP

```
MOV R1,#11H    ;R1 ← 11H
MOV R2,#22H    ;R2 ← 22H
MOV SP,#2FH    ;SP ← 2FH
PUSH R1        ;Push R1 on stack
PUSH R2        ;Push R2 on stack
POP R3         ;POP R3 from stack
POP R4         ;POP R4 from stack
```



Before MOV SP,#2FH



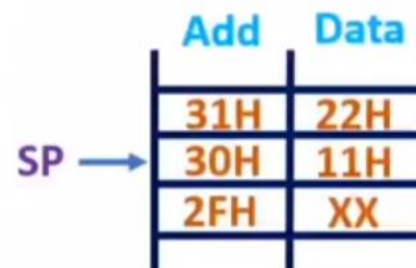
After MOV SP,#2FH



After PUSH R1



After PUSH R2



SFR of 8051

- ❑ SFR – Special Function Registers with size of 8 bits.
- ❑ SFRs are On chip registers for special functions of 8051. {Timers, Counter, IO, Serial Communication, Interrupt, Power Saving Modes, etc.}
- ❑ There are 21 SFR registers with 8051, which are used with their addresses in instructions to reduce number of opcodes in 8051.
- ❑ We have seen that internal RAM is used with addressing from 00H to 7FH. Likewise SFRs are used with addressing in between 80H to FFH.

SFR of 8051

- ❑ This addressing supports byte and bit wise operations.
- ❑ This addressing reduces numbers of opcodes for total instructions.

SETB P0.0

**;SETB has opcode & P0.0 has
address 80H**

- ❑ 8051 supports bit wise special functions and byte wise special functions, if addressing is not done with SFRs then there would be too many opcodes and that will make too much complicated instruction decode circuit.

SFR of 8051

	Name	Function	Byte Add	Bit Add
Data and Status	A	Accumulator	E0H	E7 – E0H
	B	Arithmetic {Mul. & Div.}	F0H	F7 – F0H
	PSW	Flag Register	D0H	D7 – D0H
Pointer of Memory	SP	Stack Pointer	81H	NA
	DPL	Address External Memory	82H	NA
	DPH	Address External Memory	83H	NA
IO Ports	P0	IO Latch Port	80H	87 – 80H
	P1	IO Latch Port	90H	97 – 90H
	P2	IO Latch Port	A0H	A7 – A0H
	P3	IO Latch Port	B0H	B7 – B0H
Serial Ports	SCON	Serial Port Control	98H	9F – 98H
	SBUF	Serial Port Data Buffer	99H	NA
Timer Control	TCON	Timer/Counter Control	88H	8F – 88H
	TMOD	Timer/Counter Mode Control	89H	NA
	TLO	Timer 0 Lower Byte	8AH	NA
	TL1	Timer 1 Lower Byte	8BH	NA
	TH0	Timer 0 Higher Byte	8CH	NA
Interrupt Control	TH1	Timer 1 Higher Byte	8DH	NA
	IE	Interrupt Enable	A8H	AF – A8H
	IP	Interrupt Priority	B8H	BF – B8H
Power Control	PCON	Power Control	87H	NA

Internal and External Memory

